# Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

**Q1. Which of the following statements about using approximations in physics is correct?**

**A. We can model a gymnast as a perfect sphere when predicting the torque on their elbow during a flip.**
This is incorrect because elbow torque depends on limb geometry, so a spherical model discards essential shape details.

**B. We can reasonably treat a school bus as a point mass when estimating how far it travels along a highway in 20 minutes.**
For mile-scale motion, the bus's size is negligible compared with the travel distance, so a point-mass model is appropriate for position change.

**C. We are not allowed to assume a vacuum in any problem, even when we explicitly want to neglect air drag.**
This is incorrect because "assume vacuum/no air resistance" is a standard idealization used to simplify motion models.

**D. None of the above is correct.**
This is incorrect because option B is a valid, commonly used approximation for large-scale motion.

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**Q2. Which of the following equations is correct?**

**A. $1\,\text{ms} = 1\,\text{s}$.**
This is incorrect because a millisecond is much smaller than a second.

**B. $1\,\text{ms} = 0.01\,\text{s}$.**
This is incorrect because $0.01\,\text{s}$ corresponds to $10\,\text{ms}$, not $1\,\text{ms}$.

**C. $1\,\text{ms} = 0.001\,\text{s}$.**
This is correct because the prefix "milli-" means $10^{-3}$, so $1\,\text{ms}$ equals $0.001\,\text{s}$.

**D. $1000\,\text{ms} = 0.001\,\text{s}$.**
This is incorrect because $1000\,\text{ms}$ equals $1\,\text{s}$, not $0.001\,\text{s}$.

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**Q3. Which option has the incorrect unit?**

**A. Power: watt.**
This is correct usage because the watt (W) is the SI unit of power.

**B. Momentum: kilogram·meter/second.**
This is correct usage because momentum has units of $\text{kg}\,\text{m/s}$.

**C. Pressure: newton·meter.**
This is incorrect because pressure is force per area ($\text{N/m}^2$), not $\text{N}\,\text{m}$.

**D. Electric charge: coulomb.**
This is correct usage because the coulomb (C) is the SI unit of electric charge.

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**Q4. A scout is dispatched from Winterfell, which is at the origin, and spots a group of riders at the coordinates $\vec{r} = 5\hat{\imath} - 8\hat{\jmath}$, where length units are in kiloleagues. Which of the following statements is correct?**

**A. The distance between the riders and Winterfell is 13 kiloleagues.**
This is incorrect because it treats the components as if they added linearly rather than using the Pythagorean distance.

**B. The distance between the riders and Winterfell is 8 kiloleagues.**
This is incorrect because it uses only the magnitude of the $y$-component and ignores the $x$-component.

**C. The distance between the riders and Winterfell is 5 kiloleagues.**
This is incorrect because it uses only the magnitude of the $x$-component and ignores the $y$-component.

**D. None of the above is correct.**
This is correct because the distance is $\sqrt{5^2 + (-8)^2} = \sqrt{89} \approx 9.4$ kiloleagues, which is not 13, 8, or 5.

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**Q5. If the displacement of an object is given by the function $\vec{s} = \left((0.80\,\mathrm{m/s^3})t^3 - (1.6\,\mathrm{m/s^2})t^2\right)\hat{\imath}$, which of the following statement is correct?**

**A. This is a constant acceleration motion.**
This is incorrect because differentiating twice yields an acceleration that depends on $t$, not a constant.

**B. The magnitude of the acceleration is $1.6\,\mathrm{m/s^2}$.**
This is incorrect because $1.6\,\mathrm{m/s^2}$ is a coefficient in position, not the (time-varying) acceleration.

**C. The magnitude of the acceleration at $t = 0$ is $0\,\mathrm{m/s^2}$.**
This is incorrect because the constant term in $a(t)$ is nonzero after taking two derivatives.

**D. The magnitude of the acceleration at $t = 0$ is $3.2\,\mathrm{m/s^2}$.**
This is correct because $\vec{a}(t) = \frac{d^2\vec{s}}{dt^2} = (4.8t - 3.2)\hat{\imath}$, so $|\vec{a}(0)| = 3.2\,\mathrm{m/s^2}$.

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**Q6. An object takes 4 seconds to move from position A ($\vec{r}_A = 1.0\hat{\imath} - 3.0\hat{\jmath}$) to position B ($\vec{r}_B = -5.0\hat{\imath} + 9.0\hat{\jmath}$) along a straight path, where length units are in meters. Which of the following statements is correct?**

**A. The moving distance is 13.4 m.**
This is correct because the straight-line distance is $|\vec{r}_B - \vec{r}_A| = \sqrt{(-6)^2 + (12)^2} = \sqrt{180} \approx 13.4\,\mathrm{m}$.

**B. The speed at $t = 0$ must be $3.35\,\mathrm{m/s}$.**
This is incorrect because knowing only endpoints and total time does not determine the initial speed (only the average speed).

**C. The acceleration at $t = 0$ must be $0.84\,\mathrm{m/s^2}$.**
This is incorrect because the acceleration is not specified and cannot be inferred from just two positions and a travel time.

**D. None of the above is correct.**
This is incorrect because option A correctly computes the straight-line distance.

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**Q7. A tennis ball is launched with speed $u_0$ at an angle $\phi$ above the horizontal (ignore air resistance). What is the speed of the ball when it is at the maximum height?**

**A. $u_0$.**
This is incorrect because the vertical component changes due to gravity, so the total speed generally decreases before the peak.

**B. $u_0 \cos \phi$.**
This is correct because at the top the vertical velocity is zero while the horizontal component remains $u_0 \cos \phi$, so the speed equals that value.

**C. $u_0 \sin \phi$.**
This is incorrect because $u_0 \sin \phi$ is the initial vertical component, which is not the speed at maximum height.

**D. 0.**
This is incorrect because only the vertical component becomes zero at the top, not the horizontal component.

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**Q8. A probe is traveling with an initial velocity of $\vec{v}_0 = (12.5\,\text{m/s})\hat{\imath}$ when its thrusters fire, imparting an acceleration of $\vec{a} = (-1.00\,\text{m/s}^2)\hat{\imath} + (3.50\,\text{m/s}^2)\hat{\jmath}$. Based on this scenario, which of the following statements is correct?**

**A. The x-component of the probe's velocity remains constant.**
This is incorrect because a nonzero $a_x$ changes the x-component of velocity over time.

**B. The y-component of the probe's velocity remains constant.**
This is incorrect because a nonzero $a_y$ causes the y-component of velocity to change from its initial value.

**C. Both the direction of motion and speed change.**
This is correct because acceleration changes both components of velocity, which generally changes both the velocity's direction and magnitude.

**D. None of the above is correct.**
This is incorrect because option C correctly describes the motion under a two-component acceleration.

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**Q9. If the magnitudes of both charges are doubled and their separation is cut in half, what happens to the magnitude of the electrostatic force that each charge exerts on the other?**

**A. It stays the same.**
This is incorrect because changing both charge and distance alters $q_1 q_2 / r^2$ substantially.

**B. It decreases by a factor of 4.**
This is incorrect because halving the separation increases the force (due to the $1/r^2$ dependence) rather than decreasing it.

**C. It increases by a factor of 4.**
This is incorrect because it accounts for doubling charges (factor 4) but ignores that halving distance adds another factor of 4.

**D. It increases by a factor of 16.**
This is correct because $F \propto q_1 q_2 / r^2$, so doubling both charges gives $\times 4$ and halving distance gives another $\times 4$, for a total of $\times 16$.

**E. It increases by a factor of 8.**
This is incorrect because the correct scaling combines to $4 \times 4 = 16$, not 8.

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**Q10. An object can escape a planet with radius $R$ and mass $M$ at an escape speed given by $v = \sqrt{\frac{2GM}{R}}$, where $G$ is the universal gravitational constant. Based on this expression for escape speed, which of the following statements is correct?**

**A. Doubling the planet's mass doubles the escape speed.**
This is incorrect because $v$ scales with $\sqrt{M}$, so doubling $M$ multiplies $v$ by $\sqrt{2}$, not 2.

**B. For the same planet, a heavier spacecraft needs a larger escape speed.**
This is incorrect because the formula shows escape speed depends on $M$ and $R$ of the planet, not the spacecraft's mass.

**C. For planets with the same mass, a larger radius gives a larger escape speed.**
This is incorrect because increasing $R$ decreases $M/R$, which reduces $v$.

**D. A planet with four times the mass and half the radius has a higher escape speed.**
This is correct because $v \propto \sqrt{M/R}$, so replacing $M \to 4M$ and $R \to R/2$ increases $M/R$ by a factor of 8, increasing $v$.

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**Q11. The force $\vec{F} = (-2.50\,\text{N})\hat{\imath} + (6.00\,\text{N})\hat{\jmath}$ acts on a particle as it moves along the y axis from $y = -1.2\,\text{m}$ to $y = 3.8\,\text{m}$. What is the work, in joules, done by the force on the particle?**

**A. 30.0 J.**
This is correct because $\Delta\vec{r} = (5.0\,\text{m})\hat{\jmath}$ and $W = \vec{F} \cdot \Delta\vec{r} = (6.00)(5.0) = 30.0\,\text{J}$.

**B. 17.5 J.**
This is incorrect because it comes from using the force magnitude times displacement instead of the dot product component along the motion.

**C. −12.5 J.**
This is incorrect because it incorrectly uses the x-component of force even though there is no x-displacement.

**D. 0 J.**
This is incorrect because the force has a nonzero component along the direction of motion ($\hat{\jmath}$), so the dot product is not zero.

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**Q12. A 3.0 kg sled initially moving at 4.0 m/s collides with a barrier and rebounds at a speed of 1.5 m/s moving backward. The collision lasts for 0.15 seconds. What is the average force exerted on the sled during the collision?**

**A. 110 N.**
This is correct because $\Delta p = m(v_f - v_i) = 3.0(-1.5-4.0) = -16.5\,\text{kg m/s}$, so $|F_{\text{avg}}| = |\Delta p|/\Delta t = 16.5/0.15 = 110\,\text{N}$.

**B. 73 N.**
This is incorrect because it typically results from using the wrong time interval or an incorrect velocity change.

**C. 27.5 N.**
This is incorrect because it corresponds to dividing the momentum change by 0.60 s (a common arithmetic/time misread).

**D. None of the above is correct.**
This is incorrect because option A matches the impulse-momentum calculation.

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**Q13. A 65 kg astronaut is initially at rest in space. A 0.25 kg tool is moving toward the astronaut at 18 m/s, and the astronaut catches it, holding on so they move together afterward. What is the astronaut's speed after catching the tool?**

**A. 0.069 m/s.**
This is correct because conservation of momentum for a perfectly inelastic catch gives $v_f = \frac{(0.25)(18)}{65+0.25} \approx 0.069\,\text{m/s}$.

**B. 0.32 m/s.**
This is incorrect because it comes from dividing by the astronaut's mass incorrectly or using only part of the total mass in the denominator.

**C. 18 m/s.**
This is incorrect because it ignores that momentum is shared with the astronaut, greatly reducing the final speed.

**D. 1.1 m/s.**
This is incorrect because it overestimates the result by not accounting for the astronaut's much larger mass compared with the tool.

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